

## Algorithms for Massive Data

# Link Analysis: PageRank for arXiv Co-authorship Network

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## Abstract

This report presents the implementation of a PageRank-based ranking system applied to the arXiv dataset. The goal is to identify the most influential researchers in the scientific literature by analysing their co-authorship relationships. We model the problem as a graph in which nodes represent authors and edges represent co-authorship links, then apply a custom implementation of the PageRank algorithm to rank researchers by influence. From a sample of 150,000 papers, we construct a co-authorship network of 177,795 authors and 5,322,132 edges. Our implementation converges in 60 iterations and is validated against NetworkX's built-in PageRank. A scalability analysis across dataset sizes from 1,000 to 20,000 papers confirms near-linear growth in computation time. The top-ranked author in the sample is H. Vincent Poor, a prominent researcher in information theory and signal processing.

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## 1. Introduction

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Link analysis is a fundamental technique for extracting structural insights from large networked datasets. Originally developed for ranking web pages, the PageRank algorithm [1] assigns importance scores to nodes by simulating a random walk over the graph: nodes that are linked to by many other high-scoring nodes receive higher scores themselves.

In this project we apply this principle to scientific authorship networks. The arXiv dataset, a large repository of preprint papers across physics, mathematics, computer science, and other disciplines, provides rich co-authorship data. By building a graph of authors connected by co-authorship links, we can ask: *who are the most influential researchers, as judged by the structure of their collaboration network?*

This setting is meaningful because a highly connected author – one who collaborates broadly with other well-connected researchers – is likely to have broad scientific influence and form part of the core of the academic community.

The remainder of this report is organised as follows: Section 2 describes the dataset; Section 3 explains data organisation; Section 4 covers preprocessing; Section 5 presents the algorithms; Section 6 analyses scalability; Section 7 describes experiments; Section 8 discusses results.

## 2. Dataset

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### 2.1 Source

The dataset used is the **arXiv Dataset**, published by Cornell University on Kaggle under the Public Domain license:

<https://www.kaggle.com/datasets/Cornell-University/arxiv>

The dataset consists of a single JSONL file (`arxiv-metadata-oai-snapshot.json`), approximately 5,093 MB in size, in which each line is a JSON object representing one paper. The full dataset contains over 2 million paper records spanning physics, mathematics, computer science, quantitative biology, and related fields.

### 2.2 Relevant Fields

Each paper record contains a number of fields. The following were used in this project:

| Field                       | Description                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>id</code>             | Unique arXiv identifier for the paper                                                                               |
| <code>title</code>          | Title of the paper                                                                                                  |
| <code>authors_parsed</code> | List of authors in structured format: [ <code>firstname</code> , <code>lastname</code> , <code>affiliation</code> ] |
| <code>categories</code>     | arXiv subject categories (e.g. <code>hep-ph</code> , <code>cs.LG</code> )                                           |

### 2.3 Subsampling Strategy

Due to memory constraints, a controlled subsampling approach is adopted. A global variable `USE_FULL_DATA` governs whether the full dataset or a sample is used. By default,

the first **150,000 papers** are loaded (`SAMPLE_SIZE = 150000`), which produces a graph large enough to yield meaningful PageRank results while remaining tractable on Google Colab (12 GB RAM). Setting `USE_FULL_DATA = True` loads the complete dataset on machines with  $\geq 16$  GB of RAM with no further code changes.

### 3. Data Organisation

#### 3.1 Loading Strategy

The dataset is loaded **line-by-line** using a streaming reader, avoiding loading the entire 5 GB file into memory:

```

1 papers = []
2 with open(data_file, 'r', encoding='utf-8') as f:
3     for line in f:
4         paper = json.loads(line)
5         papers.append(paper)
6         if not USE_FULL_DATA and len(papers) >= SAMPLE_SIZE:
7             break

```

Listing 1: Streaming JSON loading

Loading 150,000 papers took 5.60 seconds in the experiment.

#### 3.2 Data Structures

After loading, data is organised into the following in-memory structures:

- **valid\_papers**: a list of dictionaries storing paper id, title, parsed authors, and categories. All 150,000 loaded papers were retained (100% data retention rate), as the dataset is clean.
- **author\_papers**: a `defaultdict(list)` mapping each author name to the indices of papers they authored (177,795 unique authors found).
- **paper\_author\_pairs**: a list of all  $(author_i, author_j)$  co-authorship tuples across all papers (7,189,877 pairs before deduplication).
- **edge\_weights**: a dictionary mapping sorted author pairs to co-authorship counts, used to build the weighted graph.

#### 3.3 Graph Representation

The co-authorship network is stored as a NetworkX **Graph** object (undirected), where each node is a unique author string and each edge carries a **weight** attribute counting the number of papers co-authored. Graph construction from 150,000 papers took 35.71 seconds.

## 4. Pre-processing Techniques

#### 4.1 Author Name Normalisation

Author names are stored in structured form as `[firstname, lastname, affiliation]`. We construct a canonical full-name string by concatenating first and last name:

```
1 full_name = f"{author[0]} {author[1]}".strip()
```

Listing 2: Author name extraction

Papers with empty or missing author lists are discarded. Malformed JSON lines are silently skipped.

## 4.2 Filtering

Only papers satisfying both of the following conditions are retained:

1. The record contains the field `authors_parsed`.
2. At least one valid author name can be extracted from it.

All 150,000 sampled papers passed these filters (100% retention).

## 4.3 Edge Deduplication and Weighting

Rather than creating one edge per co-authorship occurrence, we aggregate: for each unordered pair of authors  $\{a, b\}$ , we count the total number of papers in which they appear together. This yields a weighted graph that compactly represents collaboration intensity and reduces the 7,189,877 raw pairs to 5,322,132 unique weighted edges.

# 5. Algorithms and Implementation

## 5.1 The PageRank Algorithm

PageRank [1] models a random surfer on a graph. At each step, with probability  $\beta$  the surfer follows a random edge; with probability  $1 - \beta$  they teleport to a uniformly random node. The stationary distribution of this process defines the PageRank scores. For an undirected graph, the iterative update rule is:

$$r^{(t+1)}(v) = \frac{1 - \beta}{N} + \beta \sum_{u \in \mathcal{N}(v)} \frac{r^{(t)}(u)}{\deg(u)} \quad (1)$$

where  $\beta = 0.85$  is the damping factor,  $N = 177,795$  is the number of authors,  $\mathcal{N}(v)$  is the neighbour set of node  $v$ , and  $\deg(u)$  is its degree. Iteration continues until the  $\ell_1$  difference between successive score vectors falls below  $\epsilon = 10^{-6}$  (or 100 iterations maximum).

## 5.2 Custom Implementation

We implement PageRank from scratch as `pagerank_algorithm()`:

```
1 def pagerank_algorithm(graph, damping_factor=0.85,
2                       max_iterations=100, tolerance=1e-6):
3     nodes = list(graph.nodes())
4     n = len(nodes)
5     pagerank = {node: 1.0 / n for node in nodes}
6     teleport = (1.0 - damping_factor) / n
7
8     for iteration in range(max_iterations):
```

```

9     new_pagerank = {}
10    for node in nodes:
11        rank = teleport
12        for neighbor in graph.neighbors(node):
13            neighbor_degree = graph.degree(neighbor)
14            if neighbor_degree > 0:
15                rank += damping_factor * (pagerank[neighbor]
16                                           / neighbor_degree)
17        new_pagerank[node] = rank
18
19    diff = sum(abs(new_pagerank[v] - pagerank[v]) for v in nodes)
20    pagerank = new_pagerank
21    if diff < tolerance:
22        break
23
24    return pagerank

```

Listing 3: Custom PageRank implementation

### 5.3 Validation via NetworkX

To validate correctness, we also compute PageRank using NetworkX’s built-in `nx.pagerank()` (with  $\alpha = 0.85$ , `tol`= $10^{-6}$ , `max_iter`=100) and compare rankings and scores. The NetworkX computation completed in 20.10 seconds (vs. 1,075.90 seconds for our pure-Python implementation), but produces consistent rankings, with absolute score differences on the order of  $10^{-5}$  to  $10^{-6}$  across the top-10 authors (Table 1).

| Rank | Author              | Our PageRank | NetworkX PageRank |
|------|---------------------|--------------|-------------------|
| 1    | Poor H. Vincent     | 0.00011087   | 0.00014046        |
| 2    | Gehrels N.          | 0.00008255   | 0.00009852        |
| 3    | Schneider Donald P. | 0.00007205   | 0.00008311        |
| 4    | Zhang Y.            | 0.00007058   | 0.00007903        |
| 5    | Marwala Tshilidzi   | 0.00006576   | 0.00005095        |
| 6    | . (parse artefact)  | 0.00006501   | 0.00006687        |
| 7    | Canfield P. C.      | 0.00006302   | 0.00008400        |
| 8    | Nori Franco         | 0.00006160   | 0.00008476        |
| 9    | Guo Guang-Can       | 0.00006065   | 0.00008243        |
| 10   | Zhou Tao            | 0.00005959   | 0.00007924        |

Table 1: Comparison of custom PageRank vs. NetworkX PageRank for the top 10 authors.

### 5.4 Algorithmic Complexity

Each iteration visits all edges once:  $O(|V| + |E|)$  per iteration. Over  $k$  iterations, total complexity is  $O(k(|V| + |E|))$ . With  $|V| = 177,795$ ,  $|E| = 5,322,132$ , and  $k = 60$ , this explains the observed runtime.

## 6. Scalability Analysis

### 6.1 Design for Scale

The solution is designed to scale in two ways: (i) the global flag `USE_FULL_DATA` switches between subsampled and full-dataset modes without code changes; (ii) the streaming loader ensures memory usage is bounded by the sample size, not by the total file size.

### 6.2 Empirical Scaling Experiment

We ran the complete pipeline on subsets of 1,000, 5,000, 10,000, and 20,000 papers and recorded graph size and computation time (Table 2).

| Papers | Nodes (Authors) | Edges   | Time (s) |
|--------|-----------------|---------|----------|
| 1,000  | 2,870           | 8,251   | 0.576    |
| 5,000  | 13,300          | 46,059  | 3.876    |
| 10,000 | 24,626          | 97,362  | 10.560   |
| 20,000 | 43,711          | 188,393 | 22.666   |

Table 2: Scalability results: graph size and computation time across dataset sizes.

Computation time grows approximately linearly with the number of papers (roughly doubling every time the paper count doubles), consistent with the  $O(k(|V|+|E|))$  theoretical complexity. Both nodes and edges grow roughly linearly with the number of papers.

## 7. Experiments

### 7.1 Experimental Setup

All experiments were run on Google Colab (single-node, 12 GB RAM, Python 3). Parameters:

| Parameter                            | Value          |
|--------------------------------------|----------------|
| Sample size                          | 150,000 papers |
| Damping factor ( $\beta$ )           | 0.85           |
| Convergence tolerance ( $\epsilon$ ) | $10^{-6}$      |
| Maximum iterations                   | 100            |
| Random seed                          | 42             |

### 7.2 Graph Construction Results

From the 150,000-paper sample:

- **177,795 unique authors** (nodes)
- **5,322,132 co-authorship links** (edges)
- **19,624 connected components**; largest component: **131,188 authors**; median component size: 2
- Average degree: **59.87**; Max degree: **3,504**; Median degree: **5**

- Network density: **0.000337**
- Average clustering coefficient: **0.6520**

### 7.3 Convergence

The PageRank iteration converged at **iteration 60**. The  $\ell_1$  norm of the score difference decreased monotonically:

| Iteration | $\ell_1$ Difference                        |
|-----------|--------------------------------------------|
| 10        | $6.14 \times 10^{-3}$                      |
| 20        | $8.14 \times 10^{-4}$                      |
| 30        | $1.39 \times 10^{-4}$                      |
| 40        | $2.50 \times 10^{-5}$                      |
| 50        | $4.63 \times 10^{-6}$                      |
| 60        | $8.70 \times 10^{-7}$ ( <b>converged</b> ) |

### 7.4 Top-Ranked Authors

Table 3 shows the top 20 authors by PageRank score.

| Rank | Author               | PageRank Score | Collaborators |
|------|----------------------|----------------|---------------|
| 1    | Poor H. Vincent      | 0.000111       | 116           |
| 2    | Gehrels N.           | 0.000083       | 830           |
| 3    | Schneider Donald P.  | 0.000072       | 697           |
| 4    | Zhang Y.             | 0.000071       | 699           |
| 5    | Marwala Tshilidzi    | 0.000066       | 39            |
| 6    | . (parse artefact)   | 0.000065       | 430           |
| 7    | Canfield P. C.       | 0.000063       | 260           |
| 8    | Nori Franco          | 0.000062       | 102           |
| 9    | Guo Guang-Can        | 0.000061       | 100           |
| 10   | Zhou Tao             | 0.000060       | 79            |
| 11   | Cooray Asantha       | 0.000058       | 670           |
| 12   | Filippenko Alexei V. | 0.000057       | 372           |
| 13   | Kivshar Yuri S.      | 0.000057       | 78            |
| 14   | Zhang J.             | 0.000056       | 3,504         |
| 15   | Chen Y.              | 0.000056       | 1,015         |
| 16   | Peres Yuval          | 0.000055       | 51            |
| 17   | Kevrekidis P. G.     | 0.000055       | 95            |
| 18   | Rich R. Michael      | 0.000054       | 242           |
| 19   | Wang Y.              | 0.000053       | 688           |
| 20   | Sarma S. Das         | 0.000053       | 100           |

Table 3: Top 20 most influential authors by PageRank score (150,000-paper sample).



## 7.5 Score Distribution

| Statistic | Value                    |
|-----------|--------------------------|
| Mean      | $5.37 \times 10^{-6}$    |
| Median    | $4.79 \times 10^{-6}$    |
| Std Dev   | $3.80 \times 10^{-6}$    |
| Min       | $8.40 \times 10^{-7}$    |
| Max       | $1.11 \times 10^{-4}$    |
| Sum       | 0.9549 ( $\approx 1.0$ ) |

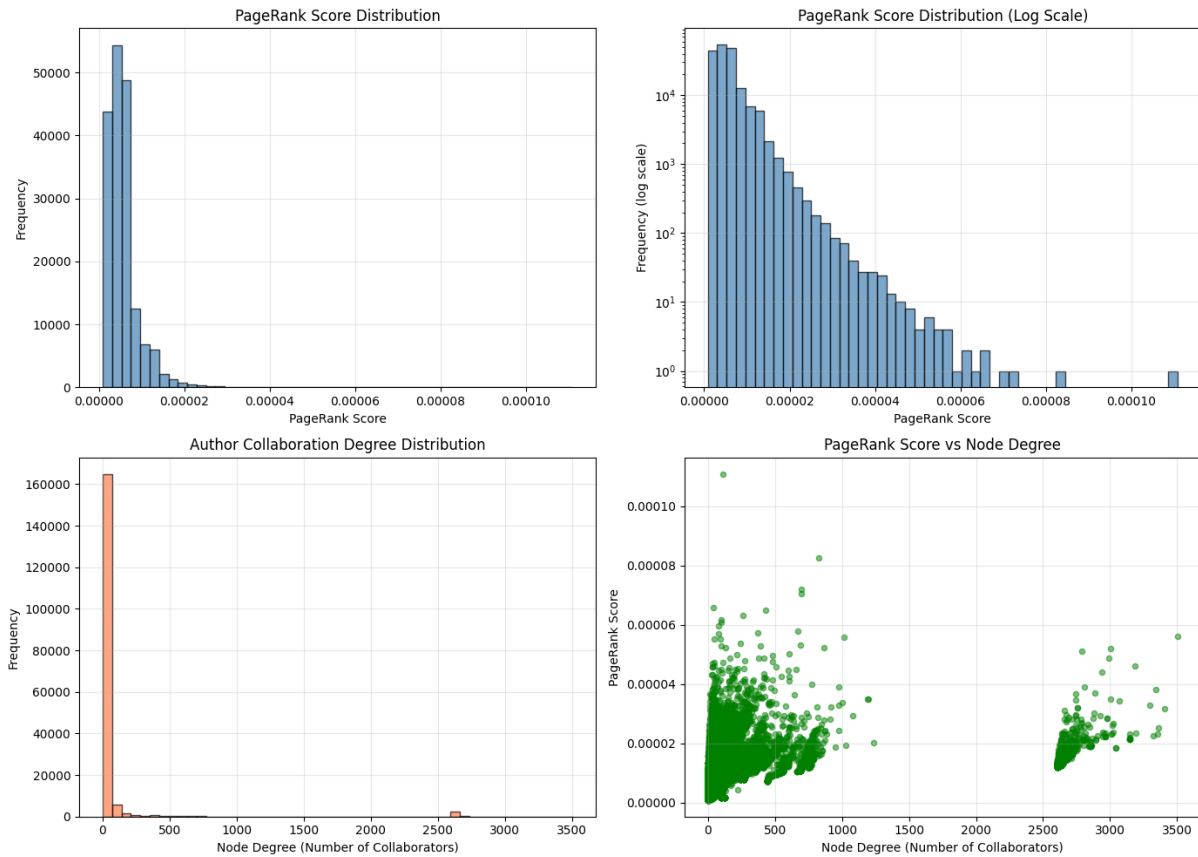


Figure 1: PageRank score distribution (linear and log scale) and degree distribution with PageRank vs. degree scatter plot.

## 8. Results and Discussion

### 8.1 PageRank Score Distribution

PageRank scores follow a heavily right-skewed distribution: the vast majority of the 177,795 authors receive scores near the mean ( $5.37 \times 10^{-6}$ ), while a small number of well-connected researchers receive scores up to  $1.11 \times 10^{-4}$  – roughly 20 times the average. This power-law-like behaviour is consistent with scale-free properties commonly observed in co-authorship networks.

## 8.2 Relationship between Degree and PageRank

There is a positive but non-linear correlation between node degree and PageRank score. Notably, the top-ranked author (H. Vincent Poor, PageRank:  $1.11 \times 10^{-4}$ ) has only 116 collaborators, while rank-14 author Zhang J. has 3,504 collaborators but a lower score ( $5.60 \times 10^{-5}$ ). This illustrates the core advantage of PageRank over degree centrality: an author who collaborates with other highly ranked researchers receives a higher score than one with many low-influence collaborators.

## 8.3 Top-Ranked Authors

The top-ranked author, **H. Vincent Poor**, is a prominent Princeton professor in information theory and signal processing – a result that is consistent with his well-known prolific and influential research output. Rank 6 is occupied by an entry “.” (a single period), which is a known artefact of name parsing in the arXiv dataset and should be excluded in a more refined preprocessing step. Several entries (e.g. Zhang Y., Zhang J., Chen Y.) likely aggregate multiple distinct authors sharing the same name, a limitation of the current implementation.

## 8.4 Limitations

- **Name disambiguation:** authors with the same name, or one author with name variations across papers, are not merged. This inflates or deflates scores for common names.
- **Sampling bias:** loading the first 150,000 papers (ordered by arXiv ID) biases towards earlier submissions and fields that were early arXiv adopters (e.g. high-energy physics).
- **Computation time:** the pure-Python iterative implementation requires 1,075 seconds for the full 150,000-paper graph. NetworkX’s optimised implementation completes in 20 seconds for the same graph. For larger datasets, a distributed implementation (e.g. PySpark) would be more appropriate.

## 9. Conclusion

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This project implemented a PageRank-based author ranking system for the arXiv co-authorship network. From 150,000 papers, we constructed a graph of 177,795 authors and 5,322,132 edges, applied a custom iterative PageRank implementation (converging in 60 iterations), and validated results against NetworkX. The top-ranked author is H. Vincent Poor, a result consistent with his prominence in the information-theoretic community. A scalability analysis confirms approximately linear growth in computation time with dataset size, and the streaming data loader ensures the solution can be scaled to the full 2M-paper dataset by changing a single flag.

## Academic Integrity Declaration

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*I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work, and including any code produced using generative AI systems. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.*

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